

House Price Prediction using Model Validation and Hyperparameter Tuning

Internship Task 3 Internship

Project Report:

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Objective:

To improve the performance of a house price prediction model by applying model validation techniques, detecting overfitting, performing cross-validation, and tuning hyperparameters using GridSearchCV to select a suitable regression model.

Dataset & Tools:

Dataset: California Housing Dataset

Libraries: pandas, NumPy, Matplotlib, Seaborn, scikit-learn

Models: Linear Regression, Ridge Regression, Decision Tree Regression

Validation: train/test split, 5-fold cross-validation, overfitting check

Tuning: GridSearchCV

Project Workflow:

1. Load and inspect the California Housing dataset.
2. Perform exploratory data analysis using distributions and a correlation heatmap.
3. Split the data into training and testing sets.
4. Standardize features for Linear and Ridge Regression.
5. Train and compare regression models using RMSE and R^2 .
6. Apply 5-fold cross-validation and check for overfitting.
7. Tune the Decision Tree with GridSearchCV.
8. Select the best-performing model and save the trained model.

Exploratory Data Analysis:

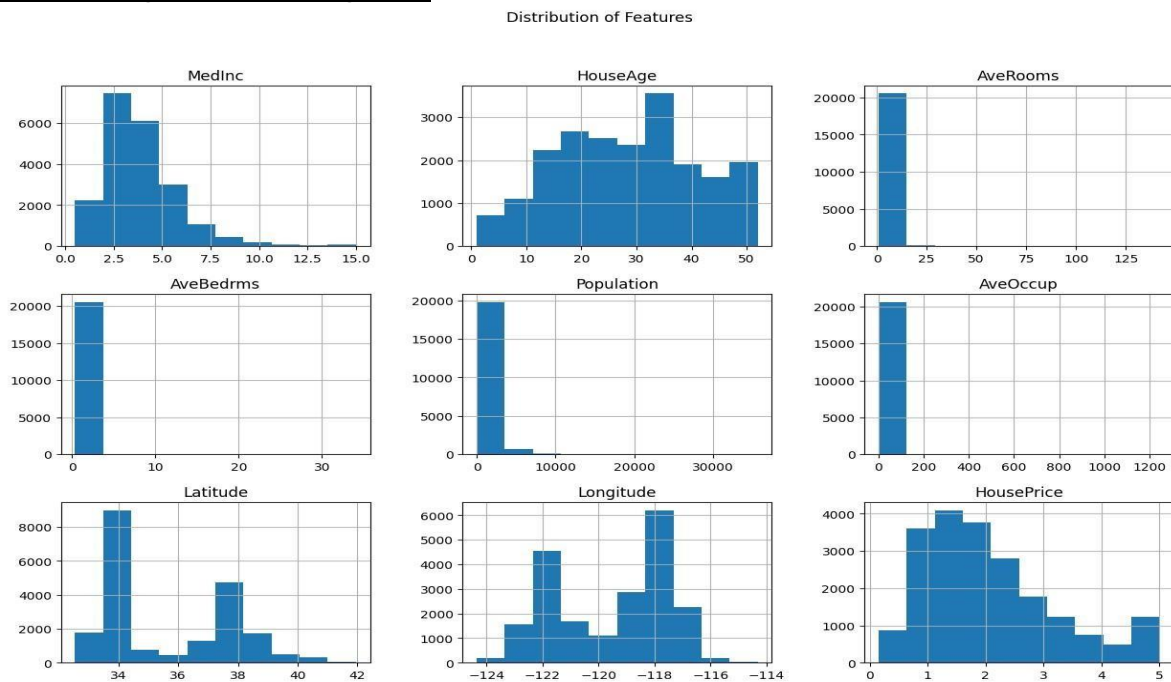


Figure 1. Feature distributions from the notebook.

Correlation Analysis and Model Comparison:

The notebook uses a correlation heatmap to examine relationships among the housing features and the target variable.

This helps identify which variables may contribute strongly to price prediction.

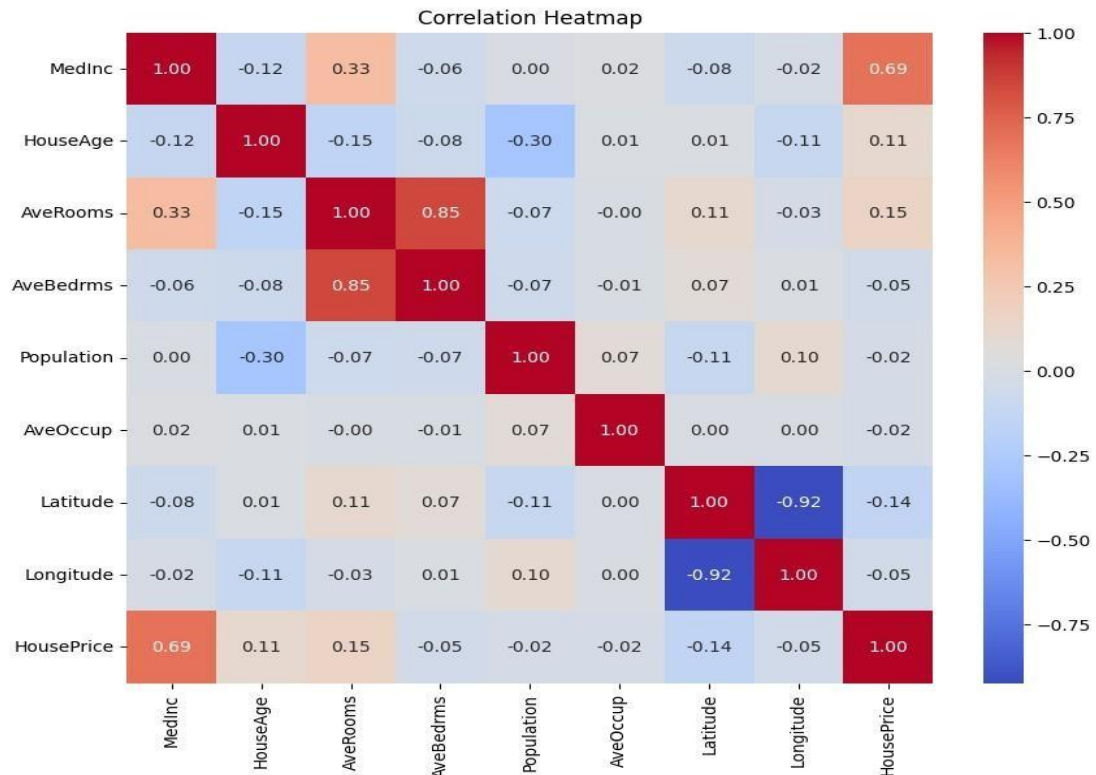


Figure 2. Correlation heatmap.

Initial Model Results:

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745557	0.575816
Decision Tree	0.649650	0.677928

The Decision Tree achieved a higher R² score and lower RMSE than the Linear and Ridge models in the initial comparison.

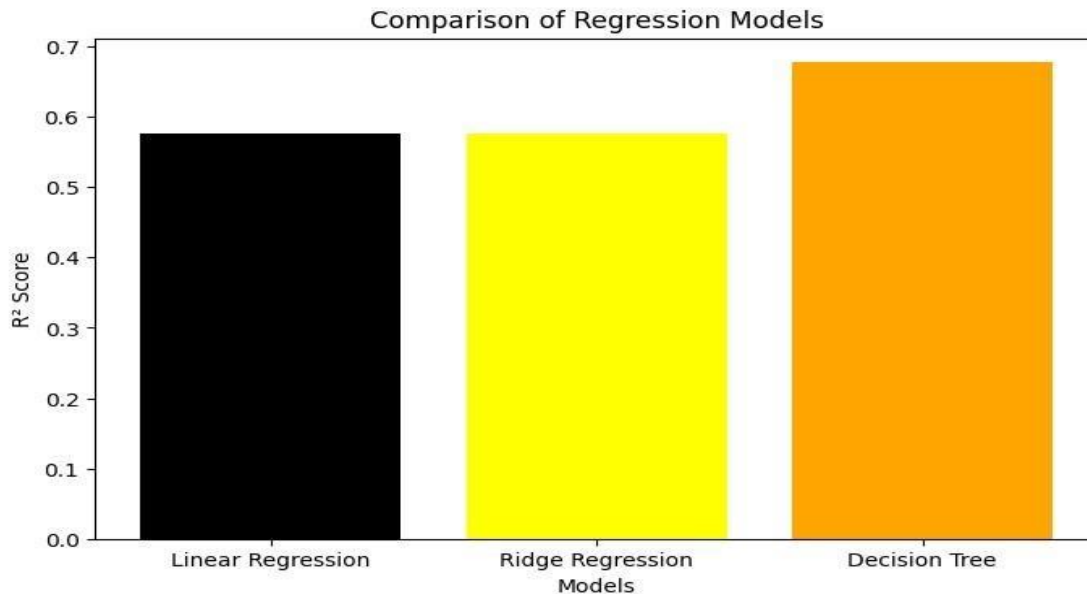


Figure 3. R² comparison of the regression models.

Model Validation, Overfitting and Hyperparameter Tuning:

Linear Regression Validation:

5-Fold Cross-Validation R²: [0.6201, 0.6130, 0.6134, 0.6107, 0.6002]

Average CV R²: 0.6115

Training R²: 0.6126 **Testing R²:** 0.5758

MAE: 0.5332 **RMSE:** 0.7456 **R²:** 0.5758

The training and testing R² difference was below the notebook's 0.10 threshold, so the notebook reported that the Linear Regression model was generalizing well.

Decision Tree Validation:

MAE: 0.4547 **RMSE:** 0.7037 **R²:** 0.6221

5-Fold CV average R²: 0.6072

Training R²: 1.0000 **Testing R²:** 0.6221

Because the Decision Tree training R² was 1.0 while the test R² was substantially lower, the notebook correctly flagged the untuned Decision Tree as overfitting.

Hyperparameter Tuning with GridSearchCV:

The notebook applies GridSearchCV with 5-fold cross-validation to tune the Decision Tree. The search varies **max_depth** over [3, 5, 7, 10, None] and **min_samples_split** over [2, 5, 10]. The best estimator is then evaluated on the test set.

Final Takeaways:

- Linear Regression provides a useful baseline.
- Cross-validation gives a more reliable estimate of generalization performance.
- The validation check shows that an unrestricted Decision Tree can overfit strongly.
- GridSearchCV is used to control tree complexity and select better hyperparameters.
- The notebook's final conclusion selects the tuned Decision Tree as the final model for the task.

Note: This report is prepared directly from the supplied Task 3 notebook. Only results that were present in the notebook outputs are reported numerically; the notebook's GridSearchCV code is summarized without inventing missing best-parameter values.